Building for the (Digital) Illiterate User

2026 // Dennie Declercq // AppDevCon

I am..

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Content

INTRO: **CITIZEN TECHNOLOGY** & **BEING SMART…**

ILLITERACY

DIGITAL ILLITERACY

BUILDING FOR THE (DIGITAL) ILLITERATE USER

Citizen Technology

- Building software for citizens, end users.
- Library app, Pharmacy website, cultural center, museum, parking app, public transport site, health tracking…
- Tools are used to make everybody’s life “better”.

Being Smart...

The Era of Smart

- We are “Smart” if we live healthy.
- Smart as customer of the right company.
- Smart if we make the “right” choices.
- Smart systems, Intelligent IT.
- In theory “smart” is linked to being intelligent.
- “The intelligent folks for the win”…

The Intelligence Quotient

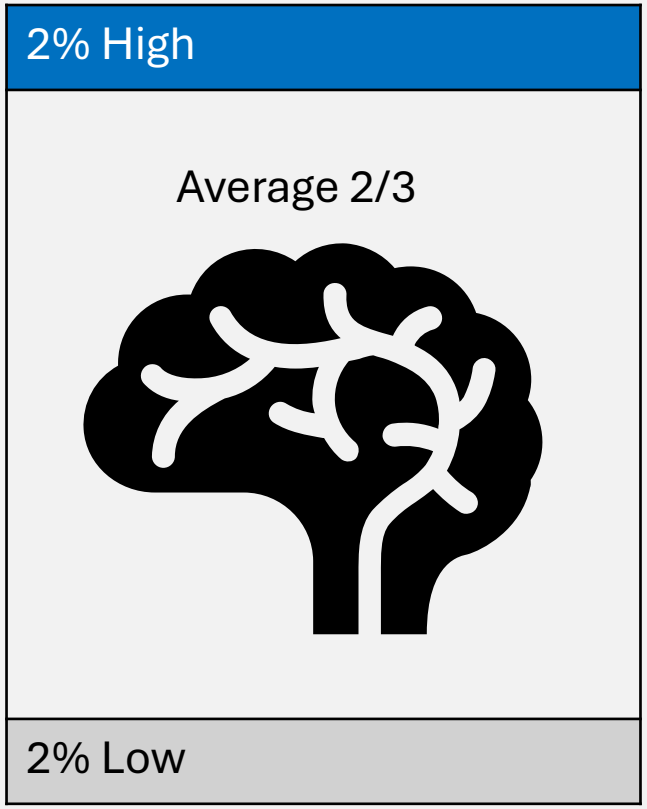
- Higher than average (2% of world population): **>130**.
- Average (2/3 of world population): **85-115**.
- Illiteracy (2% of world population): **<70**.

Not everybody is smart or intelligent!

Are smart of intelligent people more worthed?

Can we forget the “not smart” people?

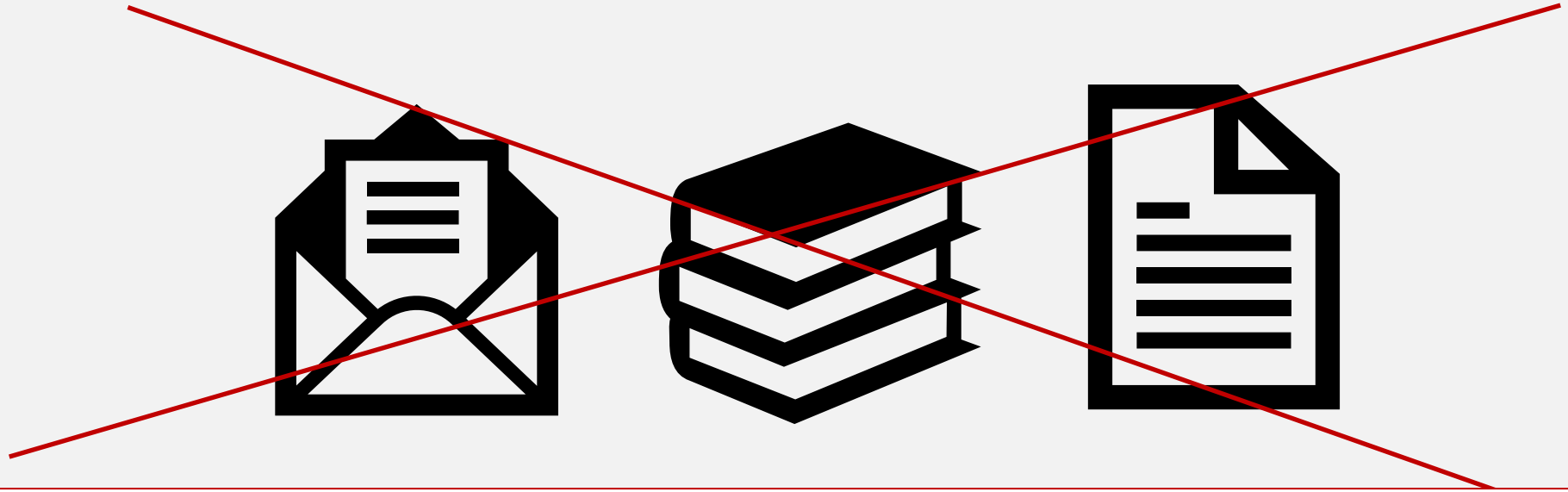
Do we need to be inclusive for the “not smart” or “illiterate” people?



Illiteracy

Not being able to read

- Not being able to read written letters, words and sentences.
- This is not only for “huge text documents”.
- Name, address, short questions,…



Not being able to write

- Not being able to **form** letters, words and sentences.
- Not being able to **write** letters, words and sentences.
- Not being able to **type** letters, words and sentences.
  - Sometimes not even their name, age, address.

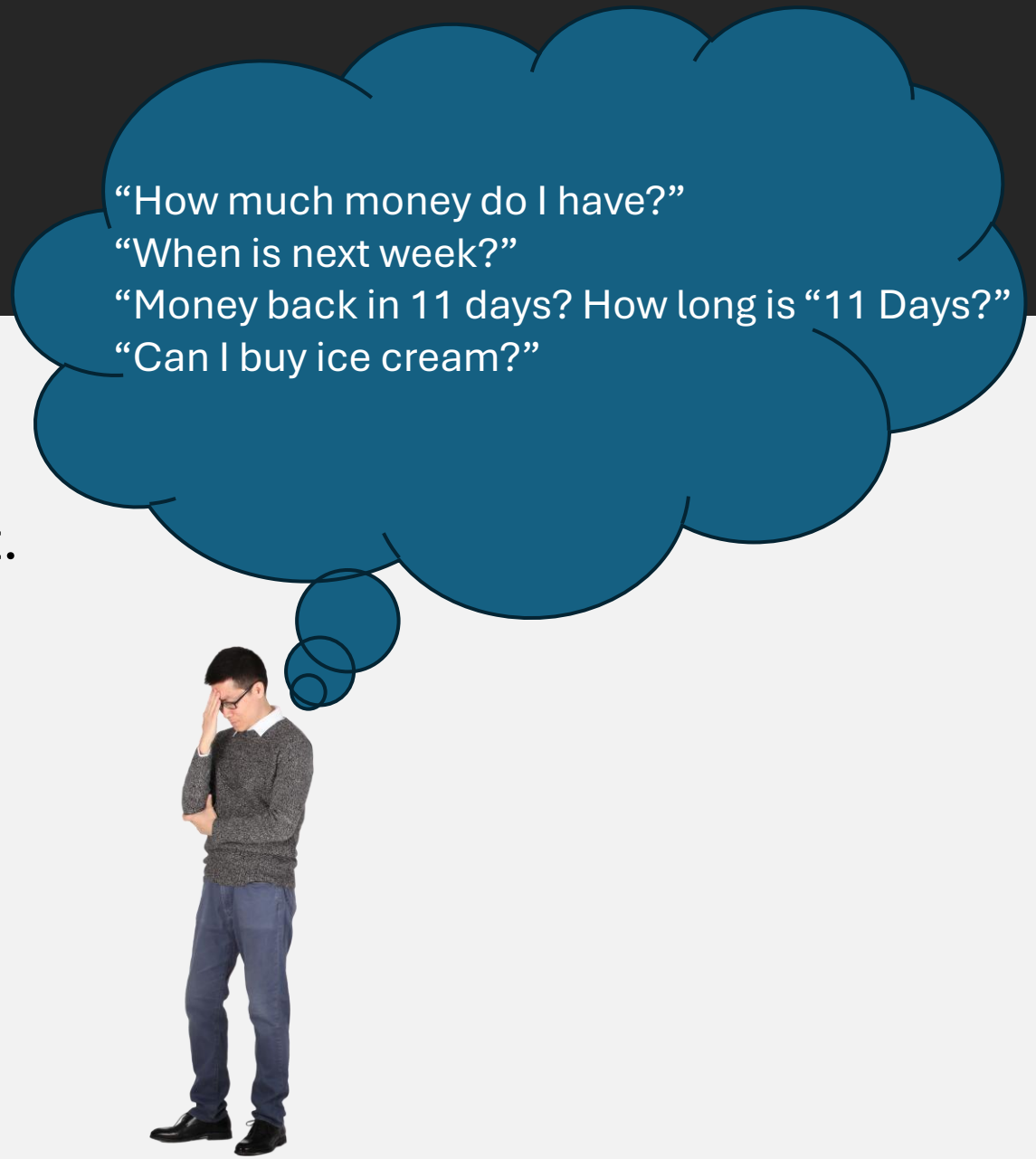
Job: ?

Job: Cimpater sjientiest

Job: Sare siver

Both affects

- General knowledge & concepts.
- Math, currency & time management.
- Personal & professional network.
- Financial possibilities.



What can help

- Pictures
- Icons
- Spoken text
- Tangibility
- Simple/ easy-to-read language

Digital illiteracy

Not familiar with digital technology

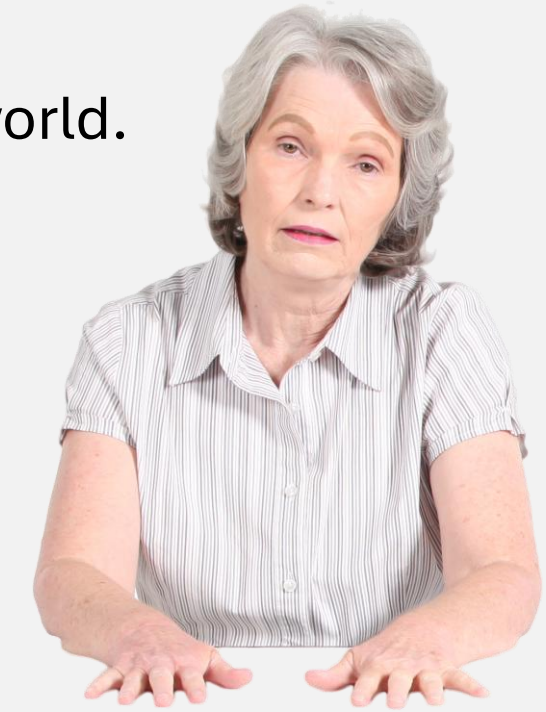
- Not grown up with digital technology.
- Lived in the physical world for more than 60-70 years without tech.
- Retired people from manual labor.
- People that never felt the need to go digital.

Not understanding the digital world

- A lot of digital concepts are in English.
- Complete new concepts that don’t have “old physical” analogies:
  - “Cloud”, “Hard Disk”, “RAM”, “CPU”

Not capable to use digital tools

- Those brand-new concepts and requirements causes barriers.
- A retired and aged brain is less capable to adjust.
- Some people can’t keep up and need to stay in the physical world.



This affects

- Not being able to drive to the city (when digital parking is mandatory).
- Not being able to enter places (if digital ticketing is required).
- Not being able to plan medical appointments and check-ups.
- Being left alone and lonely.

What can help

- Easy metaphors
- Clear instructions
- Accessible confirmations
- Human in the loop
- Physical back-up

Building for the (digital) illiterate user

Pictures and Icons

What does the following mean?

鸡肉和米饭

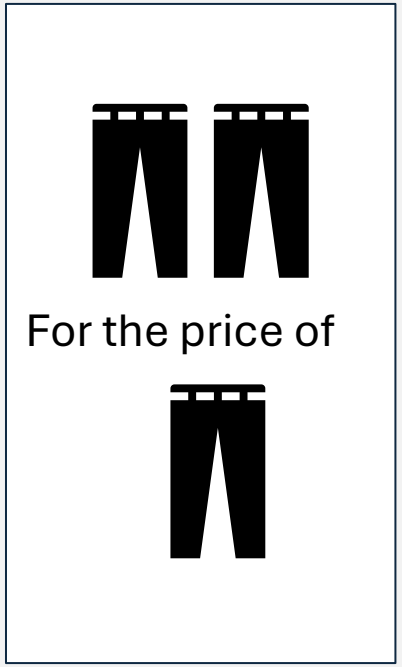
Text to Speech (Spoken text)

- Spoken may be easier to understand than written text.
- Use Text-To-Speech where technically possible.
- Text-To-Speech != a Screen reader

Welcome to TrainRide!
Here you can book your train rides.

Tangibility

- The concept of pointing to the physical world
  - Making things as easy as possible to understand;
  - Make it easy to grasp;
- Can be done with the right pictures and/or icons



Simple/ easy-to-read language

- Don’t use jargon or difficult words (Citizen Tech)
- Use simple variants of words:
- Try to limit to upper secondary vocational education.

“Welcome” vs **“accreditation accepted”**
“drinks” vs **“beverages”**
“email and password” vs **“credentials”**

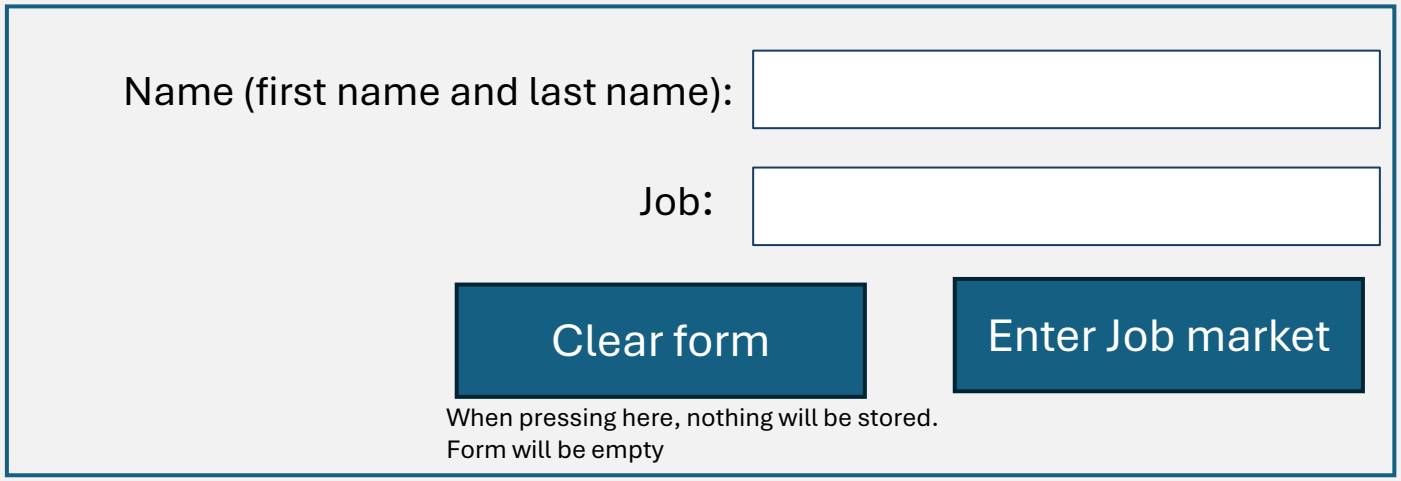
Easy metaphors

- Give metaphors in the app connecting to the physical world.
- Avoid introducing new concepts without linking to the real world.

When clicking on this button you send a mail with your order.

Clear instructions

- When asking for input, say clear what you want and in which format.
- Design the UI to avoid putting in wrong content.
- Give error messages as close to the wrong input as possible.
- When having buttons or other UI make sure to:
  - Be clear what happens if users pressed.
  - Give additional warning where needed.



Accessible confirmations

- Give instructive confirmation text when a person did an action.
- When a form is submitted, always send a copy to the users' mailbox.
- When you or the business needs to check or confirm something:
  - Give clear estimation on timing.
  - Provide visual timeline where possible.

Your request is send to our system.
Within 15 minutes you will receive a confirmation mail.
Within 2 weeks (10 business days) you will receive an answer.

Human in the loop

- Provide in person service hours.
- Train internal people and frontline workers about (digital) illiteracy.
- Be helpful and open minded.

Physical back-up

- Have a physical (ticket) shop.
- Accept Cash.
- Accept paper tickets.
- Have waiters and physical menu card.
- Provide easy to call contact center.
- Provide the option to ask for printed invoices.

Outro

Summary

- People with (digital) illiteracy exist.
- Follow training about those people.
- Be aware of designs that are accessible for those people.

Thanks // Q&A

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  Mr. #DreamingIsBelieving's Accessible Tech

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